IP Routing Questions 2

12 questions | Original English questions and source answers

Question 1

Refer to the exhibit. Router R1 Fa0/0 cannot ping router R3 Fa0/1. Which action must be taken in router R1 to help resolve the configuration issue?

```
R1 Fa0/0 10.10.10.1/24 --- Fa0/0 10.10.10.2/24 R2 Fa0/1 20.20.20.1/24 --- Fa0/1
20.20.20.2/24 R3

R1#show ip route
<output omitted>
Gateway of last resort is not set
      10.0.0.0/24 is subnetted, 1 subnets
C        10.10.10.0 is directly connected, FastEthernet0/0

R2#show ip route
<output omitted>
Gateway of last resort is not set
      20.0.0.0/24 is subnetted, 1 subnets
C        20.20.20.0 is directly connected, FastEthernet0/1
      10.0.0.0/24 is subnetted, 1 subnets
C        10.10.10.0 is directly connected, FastEthernet0/0

R3#show ip route
<output omitted>
Gateway of last resort is not set
      20.0.0.0/24 is subnetted, 1 subnets
C        20.20.20.0 is directly connected, FastEthernet0/1
      10.0.0.0/24 is subnetted, 1 subnets
S        10.10.10.0 [1/0] via 20.20.20.1
```

Options

A. configure a static route with Fa0/1 as the egress interface to reach the 20.20.20.0/24 network

B. configure a static route with 10.10.10.2 as the next hop to reach the 20.20.20.0/24 network

C. set the default gateway as 20.20.20.2

D. set the default network as 20.20.20.0/24

Answer

B. configure a static route with 10.10.10.2 as the next hop to reach the 20.20.20.0/24 network

Source: CCNA 200-301 Questions.html | Question 1

Question 2

R1 has learned route 10.10.10.0/24 via numerous routing protocols. Which route is installed?

Options

A. route with the next hop that has the highest IP

B. route with the lowest administrative distance

C. route with the lowest cost

D. route with the shortest prefix length

Answer

B. route with the lowest administrative distance

Source: CCNA 200-301 Questions.html | Question 2

Question 3

Router R1 must send all traffic without a matching routing-table entry to 192.168.1.1. Which configuration accomplishes this task?

Options

```
A. R1#config t
R1(config)#ip routing
R1(config)#ip route 0.0.0.0 0.0.0.0 192.168.1.1
```

```
B. R1#config t
R1(config)#ip routing
R1(config)#ip default-gateway 192.168.1.1
```

```
C. R1#config t
R1(config)#ip routing
R1(config)#ip route 192.168.1.1 0.0.0.0 0.0.0.0
```

```
D. R1#config t
R1(config)#ip routing
R1(config)#ip route default-route 192.168.1.1
```

Answer

A. R1#config t
R1(config)#ip routing
R1(config)#ip route 0.0.0.0 0.0.0.0 192.168.1.1

Source: CCNA 200-301 Questions.html | Question 3

Question 4

Refer to the exhibit. Router R1 is running three different routing protocols. Which route characteristic is used by the router to forward the packet that it receives for destination IP 172.16.32.1?

```
R1#show ip route
.....
D 172.16.32.0/27 [90/293847483] via 20.1.1.1
O 172.16.32.0/19 [110/22847] via 20.1.1.10
R 172.16.32.0/24 [120/2] via 20.1.1.3
```

Options

A. longest prefix

B. administrative distance

C. cost

D. metric

Answer

A. longest prefix

Source explanation

The router evaluates routes in the following order:
1. Prefix Length - The longest-matching route is preferred first. Prefix length trumps all other route attributes.
2. Administrative Distance - In the event there are multiple routes to a destination with the same prefix length, the route learned by the protocol with the lowest administrative distance is preferred.
3. Metric - In the event there are multiple routes learned by the same protocol with same prefix length, the route with the lowest metric is preferred. (If two or more of these routes have equal metrics, load balancing across them may occur.)

Source: CCNA 200-301 Questions.html | Question 4

Question 5

A packet is destined for 10.10.1.22. Which static route does the router choose to forward the packet?

Options

A. `ip route 10.10.1.0 255.255.255.240 10.10.255.1`

B. `ip route 10.10.1.20 255.255.255.252 10.10.255.1`

C. `ip route 10.10.1.20 255.255.255.254 10.10.255.1`

D. `ip route 10.10.1.16 255.255.255.252 10.10.255.1`

Answer

B. ip route 10.10.1.20 255.255.255.252 10.10.255.1

Source explanation

The subnet 10.10.1.20/30 covers the destination 10.10.1.22 as this subnet ranges from 10.10.1.20 to 10.10.1.23 so it is the correct answer.

Source: CCNA 200-301 Questions.html | Question 5

Question 6

Refer to the exhibit. Router R2 is configured with multiple routes to reach network 10.1.1.0/24 from router R1. What protocol is chosen by router R2 to reach the destination network 10.1.1.0/24?

```
R1(config)#eigrp router-id 1.1.1.1
R1(config)#network 10.1.1.0 0.0.0.255
R1(config)#network 192.168.0.1 0.0.0.0

R2(config)#interface gi1/1
R2(config-if)#ip address 192.168.0.2 255.255.255.0

R2(config)#router bgp 65001
R2(config-router)#neighbor 192.168.0.1 remote-as 65000

R2(config)#router ospf 1
R2(config)#router-id 2.2.2.2
R2(config)#network 192.168.1.2 0.0.0.0 area 0

R2(config)#router eigrp 1
R2(config)#eigrp router-id 1.1.1.1
R2(config)#network 192.168.0.1 0.0.0.0

R2(config)#ip route 10.1.1.0 255.255.255.0 192.168.0.1
```

Options

A. eBGP

B. EIGRP

C. static

D. OSPF

Answer

C. static

Source explanation

The static route has the Administrative Distance of 1 so it is preferred over other popular routing protocols.

Source: CCNA 200-301 Questions.html | Question 6

Question 7

Refer to the exhibit. How does router R1 handle traffic to 192.168.12.16?

```
EIGRP: 192.168.12.0/24
RIP: 192.168.12.0/27
OSPF: 192.168.12.0/26
```

Options

A. It selects the OSPF route because it has the lowest cost

B. It selects the EIGRP route because it has the lowest administrative distance

C. It selects the IS-IS route because it has the shortest prefix inclusive of the destination address

D. It selects the RIP route because it has the longest prefix inclusive of the destination address

Answer

D. It selects the RIP route because it has the longest prefix inclusive of the destination address

Source: CCNA 200-301 Questions.html | Question 7

Question 8

Refer to the exhibit. A router reserved these five routes from different routing information sources. Which two routes does the router install in its routing table? (Choose two)

```
IBGP route 10.0.0.0/30
RIP route 10.0.0.0/30
OSPF route 10.0.0.0/16
OSPF route 10.0.0.0/30
EIGRP route 10.0.0.1/32
```

Options

A. OSPF route 10.0.0.0/16

B. RIP route 10.0.0.0/30

C. OSPF route 10.0.0.0/30

D. EIGRP route 10.0.0.1/32

E. iBGP route 10.0.0.0/30

Answer

C. OSPF route 10.0.0.0/30

D. EIGRP route 10.0.0.1/32

Source explanation

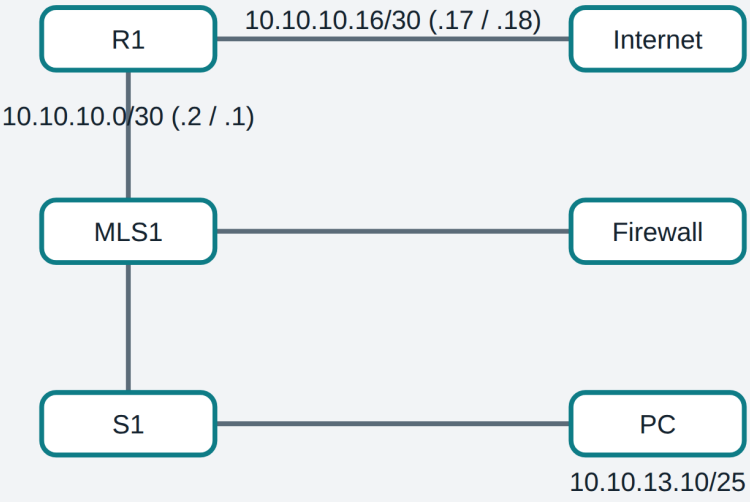
In the five above routes, there are only three different routes which are 10.0.0.0/30 and 10.0.0.0/16 and 10.0.0.1/32. Since each of these routes has a different prefix length (subnet mask), they’re considered different destinations, and they will all be installed in the routing table.

The prefix 10.0.0.0/30 was learned from three different routing protocols IBGP, RIP and OSPF so the route with lowest AD would be chosen. The AD of IBGP is 200 and the AD of RIP is 120 so for the prefix 10.0.0.0/30, only the OSPF route has the lowest AD (110) and it would be chosen to install into the routing table.

Source: CCNA 200-301 Questions.html | Question 8

Question 9

Refer to the exhibit. Which route type is configured to reach the internet?




```
R1#show ip route
Gateway of last resort is 10.10.10.18 to network 0.0.0.0
      10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C        10.10.10.0/30 is directly connected, FastEthernet0/1
O        10.10.13.0/25 [110/4576] via 10.10.10.1, 02:53:11, FastEthernet0/1
C        10.10.10.16/30 is directly connected, FastEthernet0/24
O        10.10.13.144/28 [110/110] via 10.10.10.1, 03:51:21, FastEthernet0/1
B*    0.0.0.0/0 [20/0] via 10.10.10.18, 02:17:53
```

Options

A. floating static route

B. host route

C. network route

D. default route

Answer

D. default route

Source explanation

R1 uses the last entry (B* 0.0.0.0/0 ...) to reach the Internet. The symbol * means this is a default route.

Source: CCNA 200-301 Questions.html | Question 9

Question 10

What are two reasons for an engineer to configure a floating static route? (Choose two)

Options

A. to support load balancing via static routing

B. to control the return path of traffic that is sent from the router

C. to automatically route traffic on a secondary path when the primary path goes down

D. to route traffic differently based on the source IP of the packet

E. to enable fallback static routing when the dynamic routing protocol fails

Answer

C. to automatically route traffic on a secondary path when the primary path goes down

E. to enable fallback static routing when the dynamic routing protocol fails

Source explanation

Floating static routes are static routes that have an administrative distance greater than the administrative distance (AD) of another static route or dynamic routes. By default a static route has an AD of 1 then floating static route must have the AD greater than 1. Floating static route has a manually configured administrative distance greater than that of the primary route and therefore would not be in the routing table until the primary route fails.

Source: CCNA 200-301 Questions.html | Question 10

Question 11

Refer to the exhibit. A packet is being sent across router R1 to host 172.16.3.14. To which destination does the router send the packet?

```
R1# show ip route | begin gateway
Gateway of last resort is 209.165.200.246 to network 0.0.0.0
S*    0.0.0.0/0 [1/0] via 209.165.200.246, Serial0/1/0
                  is directly connected, Serial0/1/0
      172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
S        172.16.3.0/24 [1/0] via 207.165.200.250, Serial0/0/0
O        172.16.3.0/28 [110/84437] via 207.165.200.254, 00:00:30, Serial0/0/1
      207.165.200.0/24 is variably subnetted, 6 subnets, 2 masks
C        207.165.200.244/30 is directly connected, Serial0/1/0
L        207.165.200.245/32 is directly connected, Serial0/1/0
C        207.165.200.248/30 is directly connected, Serial0/0/0
L        207.165.200.249/32 is directly connected, Serial0/0/0
C        207.165.200.252/30 is directly connected, Serial0/0/1
L        207.165.200.253/32 is directly connected, Serial0/0/1
```

Options

A. 207.165.200.254 via Serial0/0/1

B. 207.165.200.250 via Serial0/0/0

C. 207.165.200.254 via Serial0/0/0

D. 207.165.200.246 via Serial0/1/0

Answer

A. 207.165.200.254 via Serial0/0/1

Source: CCNA 200-301 Questions.html | Question 11

Question 12

Refer to the exhibit. Which command configures a floating static route to provide a backup to the primary link?

```
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
Gateway of last resort is 209.165.202.131 to network 0.0.0.0
S*    0.0.0.0/0 [1/0] via 209.165.202.131
      209.165.200.0/27 is subnetted, 1 subnets
S        209.165.200.224 [254/0] via 209.165.202.129
      209.165.201.0/27 is subnetted, 1 subnets
S        209.165.201.0 [1/0] via 209.165.202.130
```

Options

A. `ip route 209.165.200.224 255.255.255.224 209.165.202.129 254`

B. `ip route 0.0.0.0 0.0.0.0 209.165.200.224`

C. `ip route 0.0.0.0 0.0.0.0 209.165.202.131`

D. `ip route 209.165.201.0 255.255.255.224 209.165.202.130`

Answer

A. ip route 209.165.200.224 255.255.255.224 209.165.202.129 254

Source explanation

Floating static routes are static routes that have an administrative distance greater than the administrative distance (AD) of another static route or dynamic routes. By default a static route has an AD of 1 then floating static route must have the AD greater than 1. Floating static route has a manually configured administrative distance greater than that of the primary route and therefore would not be in the routing table until the primary route fails.

In this question, the floating static route to subnet 209.165.200.224/27 is manually assigned the AD of 254.

Source: CCNA 200-301 Questions.html | Question 12